

DATA SCIENCE PROJECT

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“ WINE QUALITY MACHINE LEARNING PROJECT ”

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CHAPTER - 01 INTRODUCTION

1.1

Aim :-

To develop and evaluate a Random Forest-based predictive model for assessing wine quality using physicochemical data.

The goal of this project is to :-

- Understand the key factors affecting wine quality through exploratory data analysis (EDA).
- Build a classification model to predict wine quality (Good vs. Bad) based on chemical properties such as acidity, alcohol, sulphates, etc.

Objectives :-

1. Explore and preprocess the wine quality dataset: Clean the data and prepare it for modeling.
2. Conduct exploratory data analysis (EDA): Understand the distribution and relationships of variables related to wine quality.
3. Apply feature selection: Identify key features influencing wine quality.
4. Build a Random Forest classification/regression model: Use the Random Forest algorithm to predict wine quality scores.
5. Evaluate model performance: Measure the accuracy and other relevant metrics to assess the model's effectiveness.

6. Interpret results: Analyze feature importance from the Random Forest model and provide insights into the factors affecting wine quality.

1.2

Methodology :-

Wine quality is influenced by various physicochemical properties such as acidity, sugar content, pH, and alcohol levels. To accurately predict wine quality, we performed end-to-end data analysis and modeling using Python.

Tools and Technologies Used

All data preprocessing, visualization, and machine learning modeling were implemented using the Python programming language.

Key libraries and tools used include:

- pandas → for data loading, cleaning, preprocessing, and statistical summaries
- numpy → for numerical computations
- matplotlib & seaborn → for visualizations (heatmaps, bar plots, catplots, boxplots, pairplots)
- scikit-learn (sklearn) → for train-test splitting, model building (RandomForestClassifier), and performance evaluation (accuracy, classification report, confusion matrix)

Approach

1. Data Collection:

The wine quality dataset was obtained from Kaggle, containing physicochemical attributes and quality ratings of different wine samples.

2. Data Preprocessing:

Checked for and handled missing value. Explored distributions, detected

outliers. Converted quality scores into binary classification labels (Good vs. Bad)

3. Exploratory Data Analysis (EDA):

Visualized relationships using heatmaps, catplots, and bar plots. Computed correlation coefficients to identify strong predictors of wine quality

4. Feature Selection:

Selected key features based on correlation analysis and domain knowledge. Addressed multicollinearity by analyzing highly correlated pairs

5. Model Development (Python – Random Forest):

Split data into training (80%) and testing (20%) sets using `train_test_split`. Built a `RandomForestClassifier` with 200 decision trees. (`n_estimators=200`). Trained the model on the training set and evaluated on the test set

6. Model Evaluation:

Measured accuracy, precision, recall, and F1-score. Plotted feature importances to understand the drivers of wine quality. Generated a confusion matrix to visualize classification performance

This **Python-driven methodology** ensured a reproducible, efficient, and scalable workflow from data ingestion to model interpretation. Wine quality is influenced by a variety of physicochemical properties such as acidity, sugar content, pH, and alcohol levels. Accurate prediction of wine quality based on these measurable attributes can help winemakers optimize production processes and improve product standards. In this project, we analyze a wine quality dataset using machine learning techniques, with a focus on the Random Forest model. The Random Forest algorithm is chosen due to its robustness and ability to handle complex interactions among variables. By building and evaluating this model, we aim to predict wine quality accurately and identify the most important features contributing to it. This study provides valuable insights into the factors influencing wine quality, which can be leveraged for better quality control and decision-making in the wine industry.

❖ Exploratory Data Analysis (EDA)

- Visualized relationships between variables using plots such as barplots , catplots , heatmaps etc .

- Calculated correlation coefficients to identify strong predictors of wine quality.

Heatmap — Correlations

1.2 (a) Correlation Heatmap Visualization

This heatmap shows the correlation coefficients between different physicochemical features and wine quality. The values range from -1 to 1:

- Positive correlations (closer to 1) indicate that as one feature increases, the other tends to increase.
- Negative correlations (closer to -1) mean as one feature increases, the other tends to decrease.
- Values near 0 suggest little or no linear relationship.

Pearson Correlation Coefficient Formula

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Where:

- r = Pearson correlation coefficient between variables x and y
- x_i, y_i = individual sample points of variables x and y
- $\bar{x}, \bar{y}$ = means of variables x and y
- n = number of data points

1.2 (b) What is a Catplot?

A catplot (categorical plot) is a type of visualization in seaborn designed to show the relationship between one or more categorical variables and numerical variables. It supports different kinds of plots such as boxplots, violin plots, strip plots, swarm plots, bar plots, and point plots — all useful for understanding how numerical data is distributed across categories.

Why Use Catplots for Wine Quality Data?

In your wine quality dataset, the quality score (often ranging from 3 to 8) is a discrete variable representing categories of wine quality. Meanwhile, the other features like alcohol, pH, acidity, sugar content are numerical variables. A catplot helps you:

Compare distributions of numerical features for different wine quality levels. Identify trends or patterns such as whether higher alcohol content tends to correspond with higher quality. Spot outliers or spread in numerical data across categories.

Visually summarize statistics (median, quartiles, range) per quality category.

What is a Bar Plot?

A bar plot (or bar chart) is a graphical representation that uses rectangular bars to show the quantity or value of different categories. The length or height of the bars is proportional to the value they represent. Bar plots are widely used to compare aggregated statistics (like counts, means, sums) across categories.

Why Use Bar Plots for Wine Quality Data?

In our wine quality dataset,

- **Visualize the count of wine samples per quality score** (i.e., how many wines belong to each quality category).
- **Compare average values of features like alcohol, acidity, or sugar** across different wine quality categories.
- Show **distribution and balance** of the dataset, which helps in understanding if the data is skewed toward certain quality levels.

Types of Bar Plots for Wine Data

1. Count Bar Plot:

Shows how many wine samples are in each quality category. Useful for understanding dataset balance.

2. Mean/Median Bar Plot:

Shows the average or median value of a feature (e.g., alcohol content) for each quality category. Useful for identifying trends.

Box-and-whisker plot

What is the significance of a box plot?

Box plots are a standardized way of displaying the distribution of data based on five key summary statistics:

1. Median (Q2): The middle value when the data is sorted.
2. Lower quartile (Q1): The 25th percentile — 25% of data points fall below this.
3. Upper quartile (Q3): The 75th percentile — 75% of data points fall below this.
4. Interquartile range (IQR): The range between Q1 and Q3, showing the middle 50% of data.
5. Whiskers: These typically extend to 1.5 times the IQR beyond Q1 and Q3, showing the spread of most of the data.
6. Outliers: Data points outside the whiskers are considered outliers and plotted individually.

Why is this graph useful?

- Visualize distribution: Shows the spread and skewness of the data.
- Identify outliers: Easily spot unusual values.
- Compare groups: When you have multiple box plots side-by-side, you can compare distributions across different groups or variables.
- Summary at a glance: Gives a compact summary of a dataset's central tendency and variability.

Pair plot

A pair plot (also called a scatterplot matrix) is a grid of scatterplots that shows pairwise relationships between multiple variables in a dataset. It typically includes:

- Scatterplots of every variable against every other variable.

- Histograms or KDE plots (kernel density estimates) along the diagonal to show the distribution of each variable.

In Python, you can easily create pair plots using Seaborn's `sns.pairplot()` function.

What does a pair plot show?

- Relationships between variables: Each scatterplot shows how two variables relate to each other (e.g., positive correlation, no correlation, nonlinear relationships).
 - Distributions: The diagonal plots show the distribution of each variable individually.
 - Clusters or patterns: You might spot groups or clusters in the data.
 - Outliers: Unusual points can be visually detected.
 - Potential correlations: Visual clues about variables that move together.
-

Why is it significant?

- Exploratory Data Analysis (EDA): Pair plots give a quick visual overview of relationships and distributions across many variables at once.
- Feature selection: Helps identify which variables might be strongly correlated or redundant.
- Detect patterns: Easy to spot linear or nonlinear patterns, clusters, or outliers.
- Understand dataset: Essential step before modeling or statistical testing.

Feature Selection

- Selected key features based on correlation analysis and domain knowledge.
- Reduced dimensionality to improve model accuracy and reduce overfitting.

❖ Model Development: Random Forest

- Split the dataset into training and testing sets (e.g., 70% training, 30% testing).
- Built a Random Forest model using the training data to predict wine quality.

❖ Model Evaluation

- Predicted wine quality on the test dataset.
- Evaluated model accuracy and other metrics (e.g. MAE for regression).
- Analyzed feature importance scores from the Random Forest to identify the most influential factors.

3 - 4 - 5 (content)

Regression Models

Linear Regression

Goal:

Predict a continuous outcome y based on input features x_1 , x_2 , x_3 , x_4 x_p .

Model Equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \varepsilon$$

- y : dependent variable (wine quality)
- x_j : independent variables (physicochemical features)
- β_j : coefficients (parameters to estimate)
- ε : error term (assumed to be normally distributed with mean 0)

Objective:

Find $\hat{\beta}$ to minimize the Residual Sum of Squares (RSS):

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

Performance Metrics:

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{MSE}$$

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where $\bar{y}$ is mean of observed value .

Random Forest

Random Forest is an ensemble method that builds multiple decision trees during training and outputs either the mode of classes (classification) or mean prediction (regression) of the individual trees.

Decision Tree

A decision tree partitions the feature space into regions $R_1, R_2, \dots, R_M$ and assigns a prediction value c_m for each region.

Use linear regression to establish baseline predictions of continuous wine quality scores. Use random forest regression to capture complex nonlinear interactions between chemical features and quality.

Evaluate and compare models with RMSE and R^2 .

Use Random Forest feature importance to identify which physicochemical properties most influence wine quality.

Now , The objective of this study was to develop a classification model capable of accurately predicting the target variable based on the provided features. The methodology comprised several key steps including data splitting, model selection, training, and evaluation.

Data Splitting:

The dataset was divided into training and testing subsets using the `train_test_split` function from the `sklearn.model_selection` library. An 80-20 split was applied, where 80% of the data was used for training the model, and 20% was reserved for testing its performance on unseen data. Setting the `random_state` parameter to 2 ensured that the split was reproducible, allowing consistent evaluation across multiple runs.

Model Selection and Training:

A Random Forest Classifier was selected for this task due to its robustness, ability to handle high-dimensional data, and resistance to overfitting. The classifier was instantiated with 200 decision trees (`n_estimators=200`) using the `RandomForestClassifier` class from the `sklearn.ensemble` module. The choice of 200 trees aims to balance model accuracy with computational efficiency. The model was then trained on the training subset using the `fit` method, which enabled the ensemble to learn patterns and relationships between the input features and the target labels.

Model Evaluation:

After training, the model's predictive performance was assessed on the test set. Predictions were generated using the `predict` method. The accuracy metric, computed with the `accuracy_score` function from `sklearn.metrics`, was used as the primary evaluation metric. Accuracy measures the proportion of correctly classified instances out of the total instances tested, providing a straightforward indication of the model's effectiveness. The obtained accuracy reflects the model's ability to generalize to new, unseen data, which is critical for real-world applicability.

Chapter -02 Data Overview

2.1 The dataset used in this project consists of various physicochemical properties of wine samples, along with their quality ratings and unique identifiers.

2.2 Each record represents an individual wine sample characterized by 12 variables, as described below:

- fixed acidity: Measures the concentration of non-volatile acids such as tartaric acid, which affects the wine's taste and stability.
- volatile acidity: Indicates the amount of acetic acid, which can contribute to an undesirable vinegar flavor if present in high amounts.
- citric acid: Represents the level of citric acid, contributing to the wine's freshness and flavor profile.
- Residual sugar: Quantifies the sugar remaining after fermentation, influencing sweetness.
- chlorides: Denotes the salt content in the wine.
- free sulfur dioxide: Amount of free SO_2 acting as an antimicrobial and antioxidant agent.
- total sulfur dioxide: The combined amount of free and bound SO_2 , which affects wine preservation.
- density: Reflects the wine's density, often correlated with alcohol and sugar content.
- pH: Measures the acidity or alkalinity, influencing taste and stability.
- sulphates: Level of potassium sulfate, contributing to flavor and microbial stability.
- alcohol: Percentage of alcohol content by volume.
- quality: A sensory quality score assigned by wine experts, typically ranging from 0 (lowest) to 10 (highest).
- Id: A unique identifier for each wine sample.

2.3 Let's have a look on dataset

Fixed acidity	Volatile acidity	Citric acid	Residual sugar	chlorides	Free Sulfur dioxide	total sulfur dioxide	Density	pH	Sulphates	Alcohol	Quality	Id
7.4	0.7	0	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5	0
7.8	0.88	0	2.6	0.098	25	67	0.9968	3.2	0.68	9.8	5	1
7.8	0.76	0.04	2.3	0.092	15	54	0.997	3.26	0.65	9.8	5	2
11.2	0.28	0.56	1.9	0.075	17	60	0.998	3.16	0.58	9.8	6	3
7.4	0.7	0	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5	4
7.4	0.66	0	1.8	0.075	13	40	0.9978	3.51	0.56	9.4	5	5
7.9	0.6	0.06	1.6	0.069	15	59	0.9964	3.3	0.46	9.4	5	6
7.3	0.65	0	1.2	0.065	15	21	0.9946	3.39	0.47	10	7	7
7.8	0.58	0.02	2	0.073	9	18	0.9968	3.36	0.57	9.5	7	8
6.7	0.58	0.08	1.8	0.097	15	65	0.9959	3.28	0.54	9.2	5	10
5.6	0.615	0	1.6	0.089	16	59	0.9943	3.58	0.52	9.9	5	12
7.8	0.61	0.29	1.6	0.114	9	29	0.9974	3.26	1.56	9.1	5	13
8.5	0.28	0.56	1.8	0.092	35	103	0.9969	3.3	0.75	10.5	7	16
7.9	0.32	0.51	1.8	0.341	17	56	0.9969	3.04	1.08	9.2	6	19
7.6	0.39	0.31	2.3	0.082	23	71	0.9982	3.52	0.65	9.7	5	21
7.9	0.43	0.21	1.6	0.106	10	37	0.9966	3.17	0.91	9.5	5	22
8.5	0.49	0.11	2.3	0.084	9	67	0.9968	3.17	0.53	9.4	5	23
6.9	0.4	0.14	2.4	0.085	21	40	0.9968	3.43	0.63	9.7	6	24
6.3	0.39	0.16	1.4	0.08	11	23	0.9955	3.34	0.56	9.3	5	25
7.6	0.41	0.24	1.8	0.08	4	11	0.9962	3.28	0.59	9.5	5	26
7.1	0.71	0	1.9	0.08	14	35	0.9972	3.47	0.55	9.4	5	28
7.8	0.645	0	2	0.082	8	16	0.9964	3.38	0.59	9.8	6	29
6.7	0.675	0.07	2.4	0.089	17	82	0.9958	3.35	0.54	10.1	5	30
8.3	0.655	0.12	2.3	0.083	15	113	0.9966	3.17	0.66	9.8	5	32
5.2	0.32	0.25	1.8	0.103	13	50	0.9957	3.38	0.55	9.2	5	34
7.8	0.645	0	5.5	0.086	5	18	0.9986	3.4	0.55	9.6	6	35
7.8	0.6	0.14	2.4	0.086	3	15	0.9975	3.42	0.6	10.8	6	36
8.1	0.38	0.28	2.1	0.066	13	30	0.9968	3.23	0.73	9.7	7	37
7.3	0.45	0.36	5.9	0.074	12	87	0.9978	3.33	0.83	10.5	5	40
8.8	0.61	0.3	2.8	0.088	17	46	0.9976	3.26	0.51	9.3	4	41
7.5	0.49	0.2	2.6	0.332	8	14	0.9968	3.21	0.9	10.5	6	42
8.1	0.66	0.22	2.2	0.069	9	23	0.9968	3.3	1.2	10.3	5	43
4.6	0.52	0.15	2.1	0.054	8	65	0.9934	3.9	0.56	13.1	4	45
7.7	0.935	0.43	2.2	0.114	22	114	0.997	3.25	0.73	9.2	5	46

TOTAL COUNT = 1144

Chapter - 03 Exploratory Data Analysis

3.1 Descriptive statistics

3.1.1

Code :-

```
import pandas as pd
data=pd.read_csv("/content/WineQT.csv")
data.describe()
```

Summary Statistics of Wine Dataset								
Feature	Count	Mean	Std Dev	Min	25%	50%	75%	Max
fixed acidity	1143	8.31	1.75	4.60	7.10	7.90	9.10	15.90
volatile acidity	1143	0.531	0.179	0.12	0.393	0.520	0.640	1.58
citric acid	1143	0.268	0.197	0.00	0.090	0.250	0.420	1.00
residual sugar	1143	2.53	1.36	0.60	1.90	2.00	2.60	15.50
chlorides	1143	0.087	0.047	0.012	0.070	0.079	0.090	0.611
free sulfur dioxide	1143	15.62	10.25	1.00	7.00	13.00	21.00	68.00
total sulfur dioxide	1143	45.91	32.78	6.00	21.00	37.00	61.00	289.00
density	1143	0.9967	0.0019	0.990	0.995	0.996	0.997	1.003
pH	1143	3.31	0.157	2.74	3.21	3.31	3.40	4.01
sulphates	1143	0.658	0.170	0.33	0.55	0.62	0.73	2.00
alcohol	1143	10.44	1.08	8.40	9.50	10.20	11.10	14.90
quality	1143	5.66	0.81	3.00	5.00	6.00	6.00	8.00
Id	1143	804.97	463.99	0.00	411.00	794.00	1209.50	1597.00

Key note :

The average wine quality is around 5.66, with most wines scoring between 5 and 6. Alcohol content ranges from 8.4% to 14.9%, with higher alcohol levels often associated with better quality.

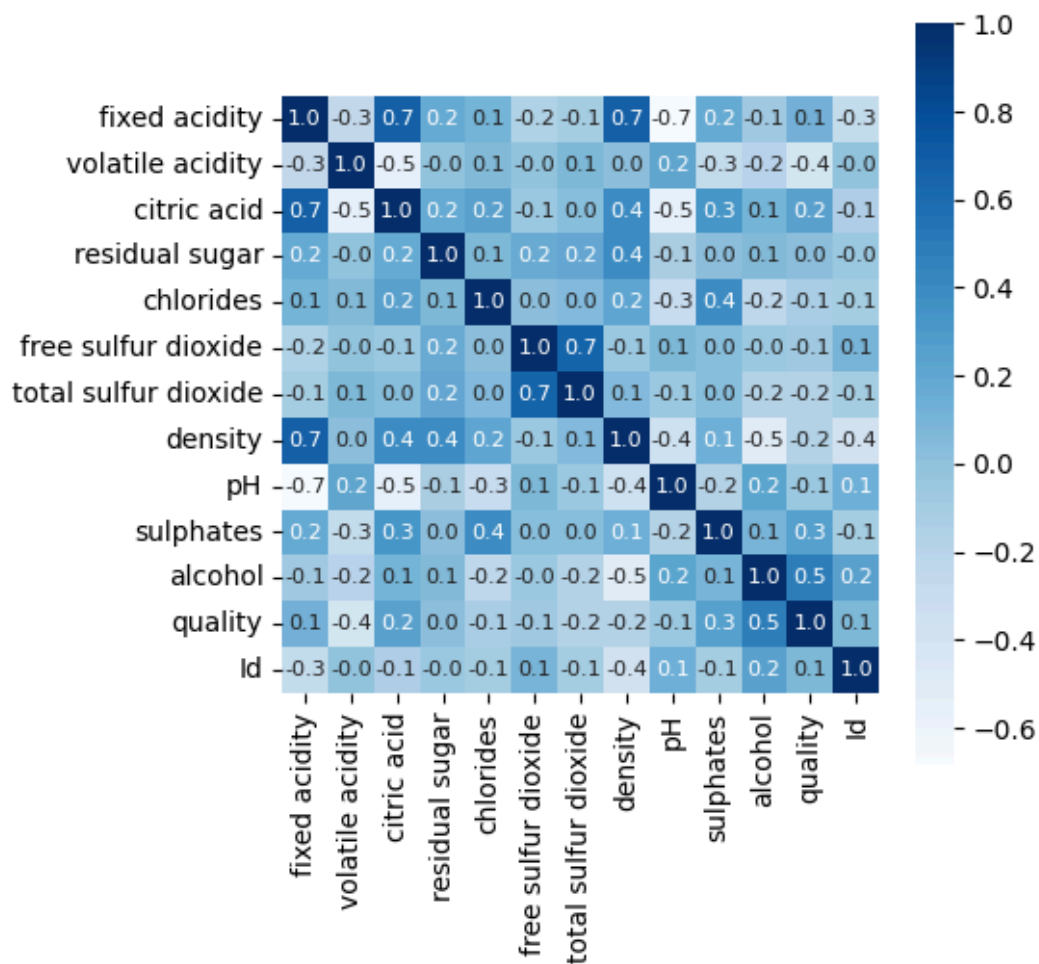
Fixed acidity, residual sugar, and sulphates vary across samples and may significantly affect taste and quality.

Some features like citric acid and chlorides have relatively low mean values and tight distributions.

3.1.2 Heat Map for Correlations

Code :

```
import matplotlib.pyplot as plt
import seaborn as sns
correlation = data.corr()
plt.figure(figsize=(5,5))
sns.heatmap(correlation,cbar=True,square=True,fmt='.1f',annot=True,anno
t_kws={'size':8},cmap='Blues')
```



Relation to Target Variable (quality**):**

- Alcohol (0.5): Strongest positive correlation with wine quality
→ higher alcohol = better quality.

- Volatile Acidity (-0.4): Moderate negative correlation → higher volatility = lower quality.
- Sulphates (0.3): Mild positive correlation with quality.
- Citric Acid (0.2): Slightly positive correlation → may enhance freshness and quality.
- Other features (e.g., residual sugar, chlorides, sulfur dioxide) have weak or negligible correlation with quality.

❖ **Highly Correlated Features (Multicollinearity Risk):**

- Fixed Acidity & Citric Acid (0.7): Strong positive correlation → both contribute to acidity.
- Free & Total Sulfur Dioxide (0.7): As expected, total SO₂ includes free SO₂.
- Density & Alcohol (-0.5): Negative correlation → higher alcohol decreases density.
- Density & Residual Sugar (0.4): Moderate positive correlation → more sugar increases density.

❖ **Weakly Related or Redundant Features:**

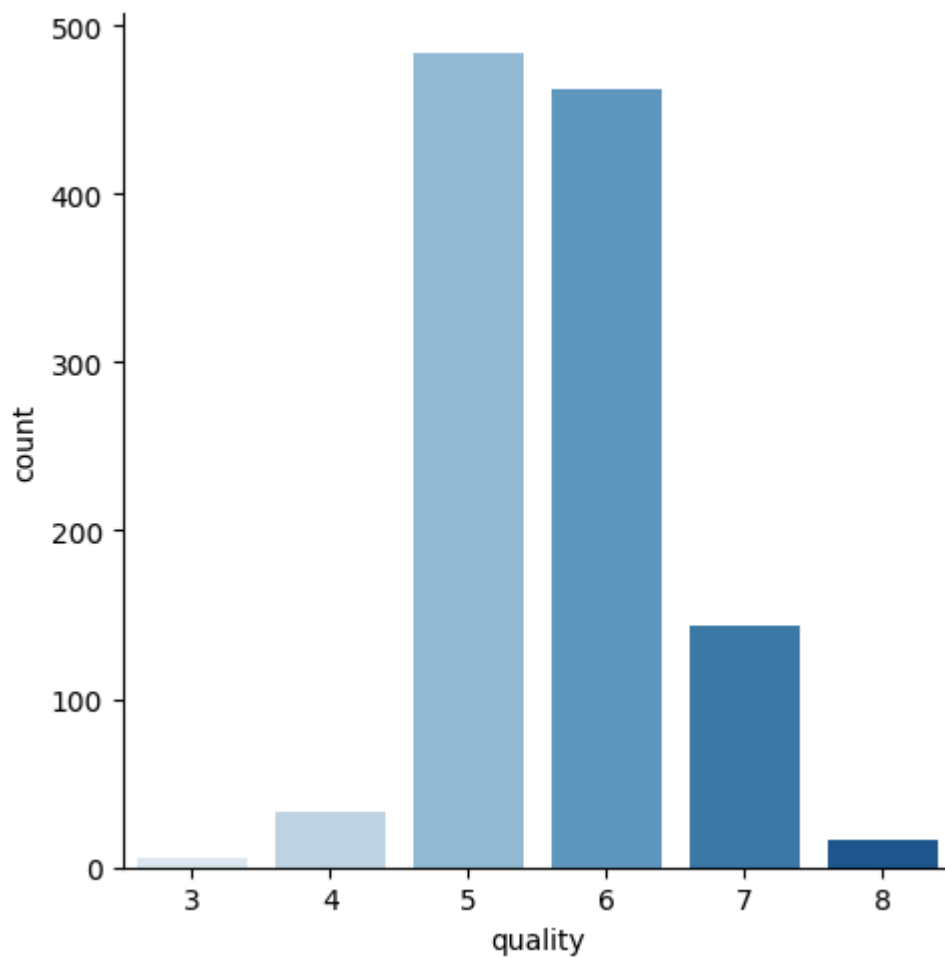
- Chlorides, residual sugar, Id: Very low correlation with quality → limited predictive value individually.

3.1.3 Catplot (check quality) and Barplot

Code :-

```
plt.figure(figsize=(5,5))  
  
sns.catplot(x="quality",data=data,kind="count",palette="Blues")  
  
#large no of samples for low quality lies in 5 6
```

Graph :- Between count and quality

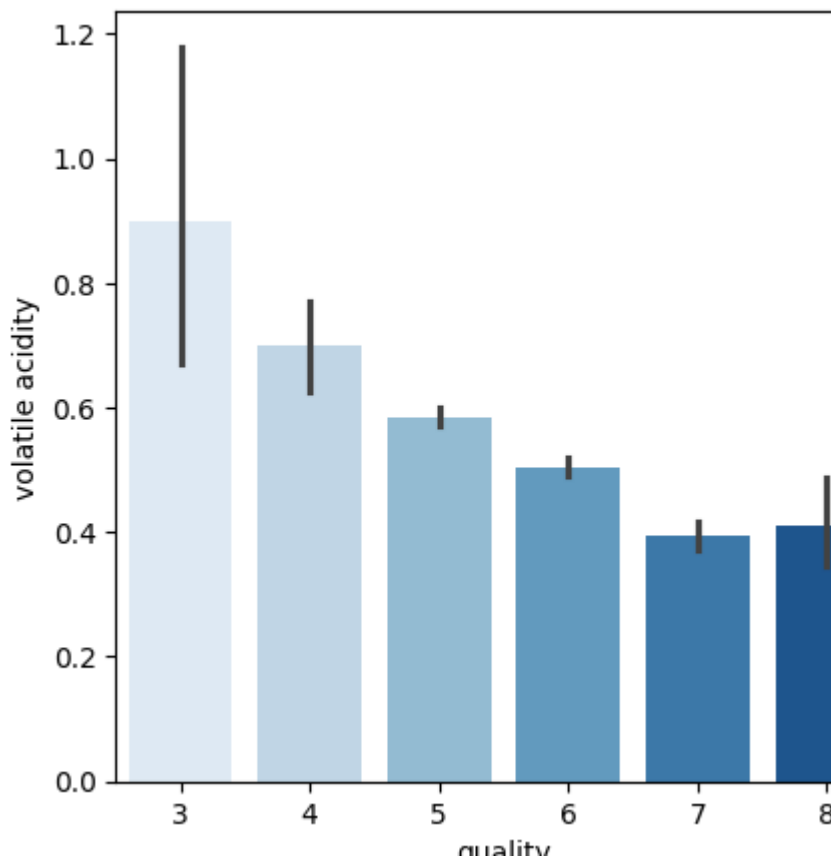


The majority of wines in the dataset have a quality rating of 5 or 6, indicating that most wines are of average quality.

Code :- Barplot

```
plt.figure(figsize=(5,5))  
sns.barplot(x='quality',y='volatile acidity',data=data,palette='Blues')
```

Graph between Quality and Volatile Acidity

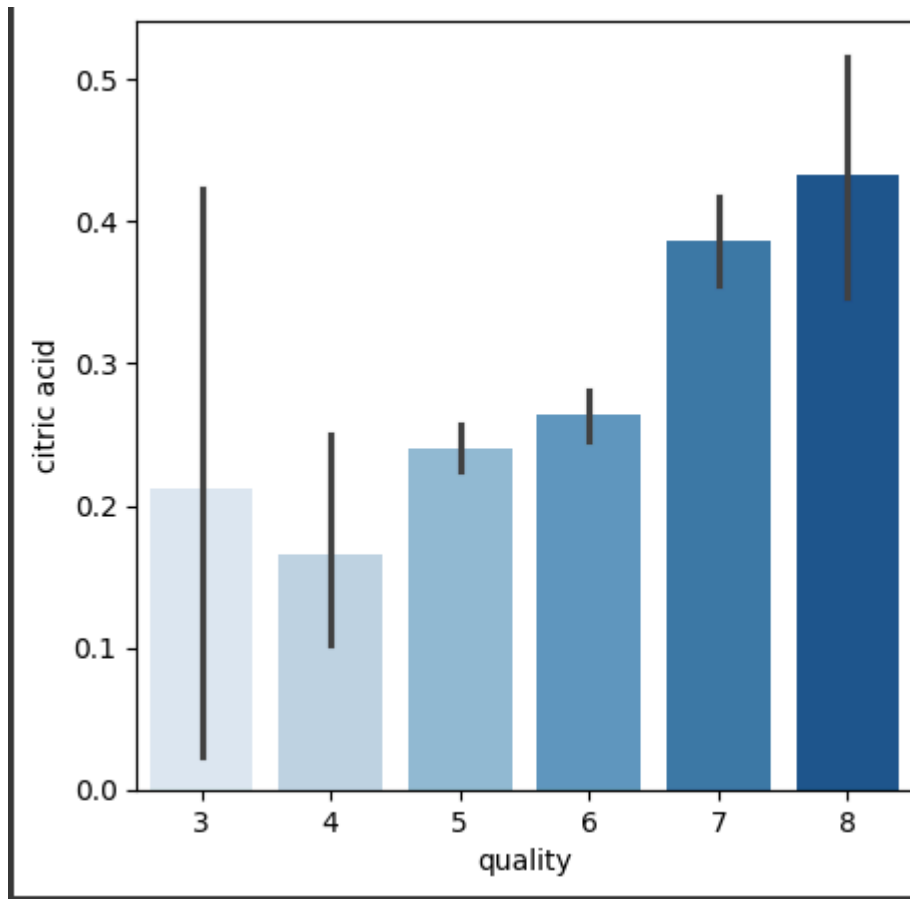


There is a clear negative trend between volatile acidity and wine quality. Lower quality wines (quality 3 and 4) have the highest average volatile acidity levels. As wine quality increases from 5 to 8, the average volatile acidity decreases steadily.

Graph between Quality and citric acid

Code :-

```
plt.figure(figsize=(5,5))  
sns.barplot(x='quality',y='citric acid',data=data,palette='Blues')
```



There is a positive relationship between citric acid content and wine quality. Higher quality wines (quality 7 and 8) tend to have higher average citric acid levels. Lower quality wines (3 and 4) have relatively lower citric acid content. This suggests that citric acid may contribute to perceived wine freshness and quality.

Box and whisker plot

Code :-

```
import matplotlib.pyplot as plt
import seaborn as sns

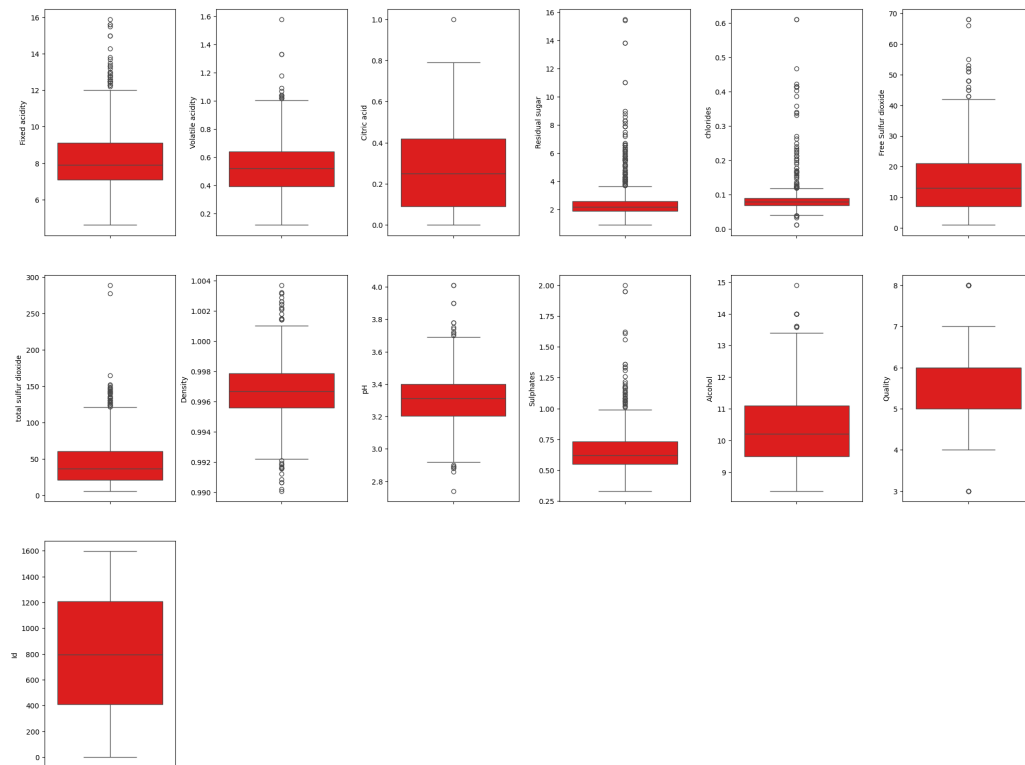
num_plots = len(data.columns) # number of columns
ncols = 6
nrows = (num_plots + ncols - 1) // ncols # ceil division to get rows

fig, ax = plt.subplots(nrows=nrows, ncols=ncols, figsize=(20, nrows*5))
ax = ax.flatten()

for index, col in enumerate(data.columns):
    sns.boxplot(y=col, data=data, color='r', ax=ax[index])

# Hide any unused subplots if number of columns < nrows*ncols
for i in range(index + 1, len(ax)):
    fig.delaxes(ax[i])

plt.tight_layout(pad=0.5, w_pad=0.7, h_pad=5.0)
plt.show()
```



Variable-wise insights:

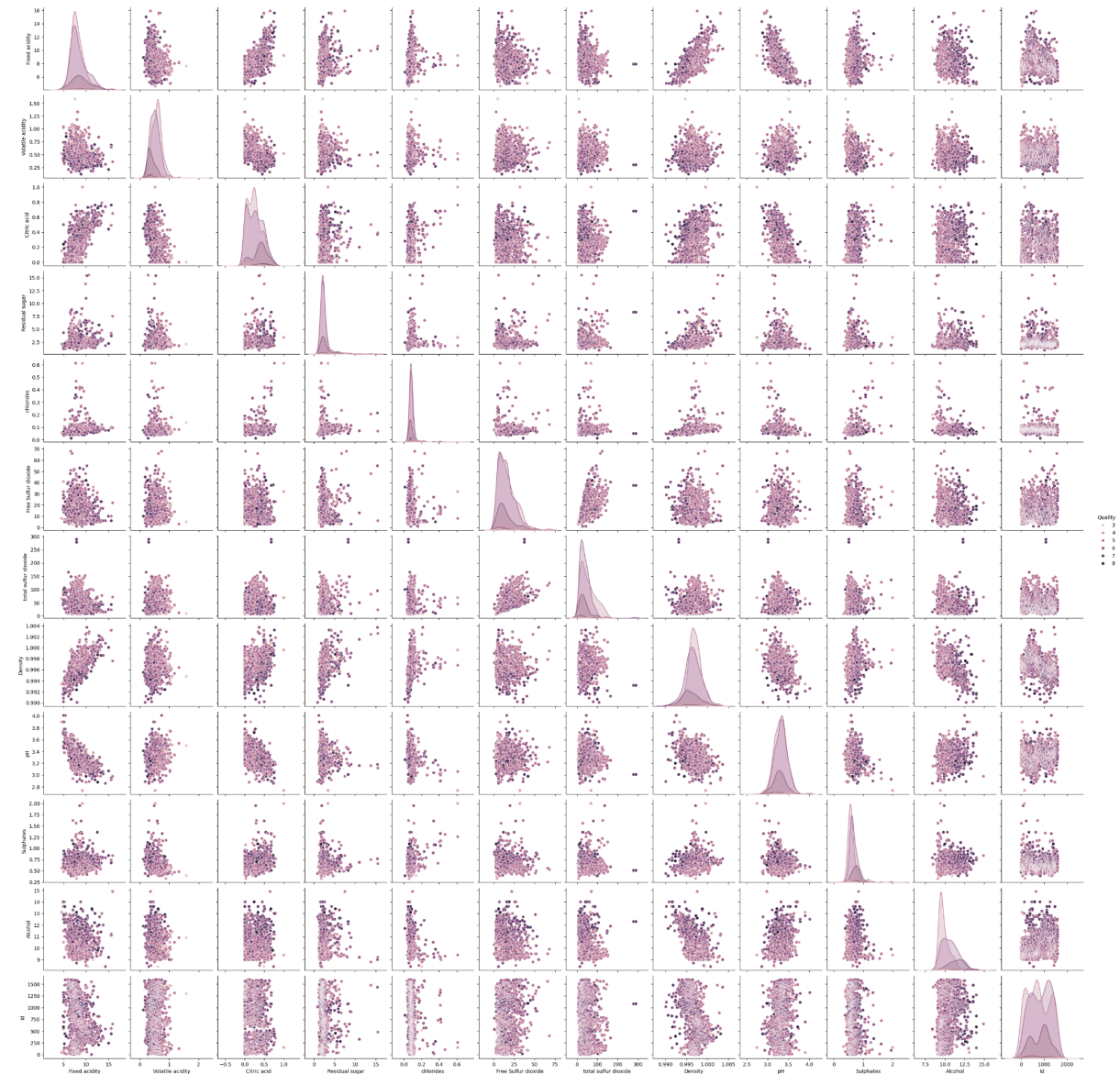
- Fixed Acidity**
 Median around 8. Some outliers with very high acidity (above ~12).
 This means most wines have acidity between ~7 and 9, but a few have notably higher acidity.
- Volatile Acidity**
 Median around 0.5. Few outliers slightly above 1.0.
 Volatile acidity is mostly low and concentrated, with a few wines having higher values.
- Citric Acid**
 Median around 0.25. Skewed slightly right; a few samples have higher citric acid. Most wines have low to moderate citric acid content.
- Residual Sugar**
 Median around 2. Several outliers with very high sugar (up to ~15).
 Suggests most wines are dry, but some have much higher sugar content.
- Chlorides**
 Median below 0.1. Outliers extending above 0.3. Most wines have low chloride levels.

6. Free Sulfur Dioxide
Median around 15. Large spread with some extreme high outliers (up to ~70).
Shows variation in preservative levels among wines.
7. Total Sulfur Dioxide
Median near 35. Several extreme outliers (up to almost 300).
Indicates variability in total preservative content, some very high.
8. Density
Very tight distribution around 0.996 to 0.998. Very few outliers.
Density doesn't vary much among samples.
9. pH
Median around 3.3. Some outliers at lower and higher ends. Most wines have slightly acidic pH around 3.2-3.4.
10. Sulphates
Median around 0.65. Several outliers above 1.0. Sulphate levels vary, but mostly concentrated around 0.5-0.7.
11. Alcohol
Median around 10.
Outliers on the higher end (up to 14-15).
Alcohol content is mostly between 9 and 12, with some stronger wines.
12. Quality
Median around 5-6. Some variation between 3 and 8.
Shows the quality ratings spread in the dataset.
13. Id
Wide range, showing that IDs are just an index, not a meaningful variable for analysis.

Pair plot

Code :-

```
sns.pairplot(data, hue='Quality')
```



Strong positive correlations:

Free Sulfur Dioxide vs Total Sulfur Dioxide: As expected, these are positively correlated. Fixed Acidity vs Citric Acid: Some positive correlation visible.

Negative correlations:

pH vs Fixed Acidity: When acidity increases, pH decreases (more acidic means lower pH).

Clusters by Quality:

Higher quality wines (darker points) tend to be associated with higher alcohol content.

Other variables show less clear separation by quality in these scatterplots, but subtle trends may exist.

No obvious correlation:

Variables like Chlorides and Residual Sugar show more scattered points without clear linear relationships.

Distribution plot

Code #Dist plot of all features:

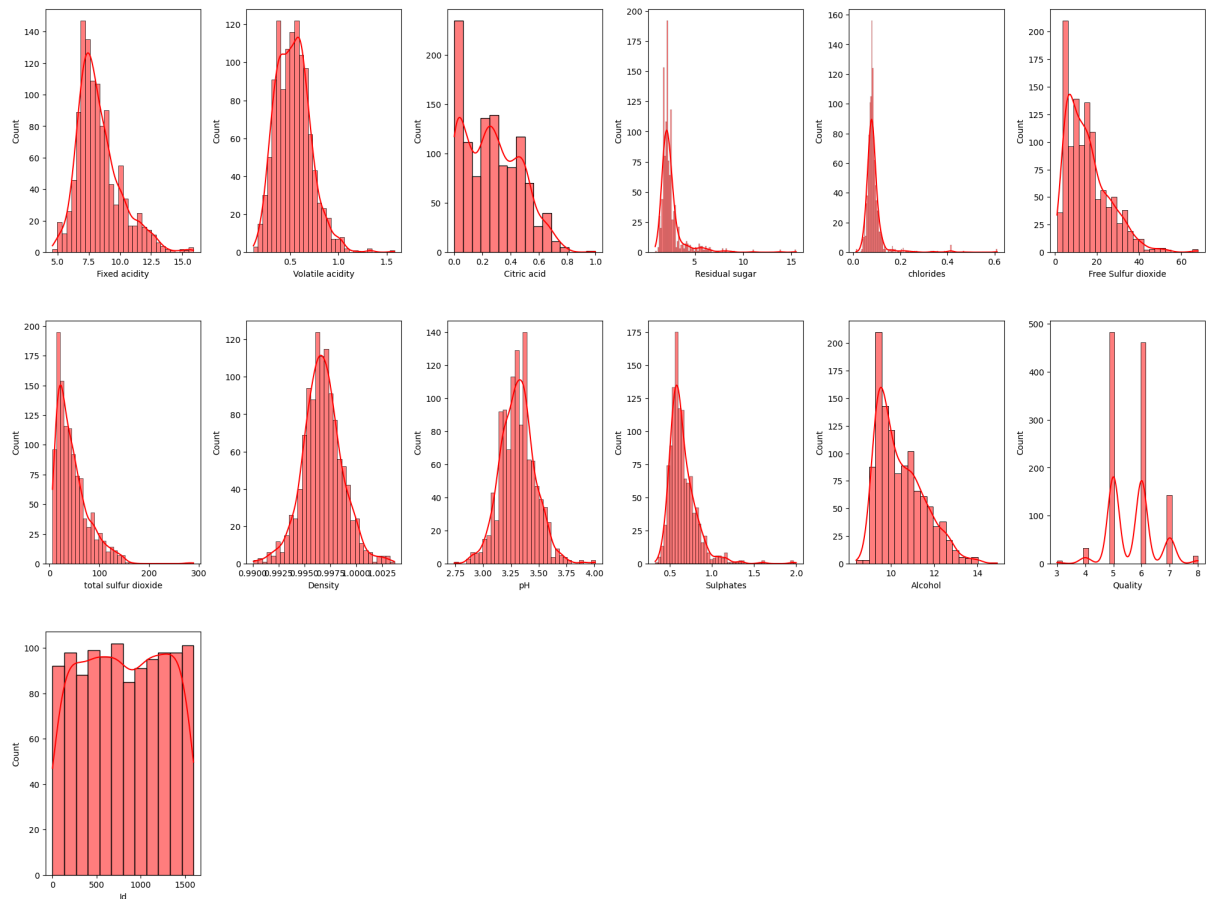
```
# create dist plot
n = len(data.columns)
cols = 6
rows = n // cols + int(n % cols != 0) # Calculate rows needed

fig, ax = plt.subplots(nrows=rows, ncols=cols, figsize=(20, rows * 5))
ax = ax.flatten()

for index, col in enumerate(data.columns):
    sns.histplot(data[col], color='r', ax=ax[index], kde=True)

# Hide any unused subplots
for i in range(index + 1, len(ax)):
    fig.delaxes(ax[i])

plt.tight_layout(pad=0.5, w_pad=0.7, h_pad=5.0)
plt.show()
```



Explanation for Each Plot:

1. Fixed acidity:

Most values are clustered around 7-9. Distribution is somewhat normal but slightly skewed right.

2. Volatile acidity:

Most data points are between 0.3 and 1. Looks fairly normally distributed but skewed right.

3. Citric acid:

Concentrated near zero with a spread toward 0.5. Bimodal or multi-peak structure (two or more peaks).

4. Residual sugar:

Right skewed (most values are low, few high). Lots of wines with low residual sugar.

5. Chlorides:

Very right skewed; most data points are near zero.

6. Free sulfur dioxide:

Skewed to the right; majority have low values, but some have much higher values.

7. Total sulfur dioxide:

Similar to free sulfur dioxide but spread wider.
Right skewed distribution.

8. Density:

Very narrow range with a bell curve. Density values are tightly clustered.

9. pH:

Normally distributed, centered around 3.2 to 3.4.

10. Sulphates:

Right skewed; most values below 1.

11. Alcohol:

Right skewed, with most values between 9-12%.

12. Quality:

Discrete variable (probably integer quality scores). Shows multiple peaks reflecting different wine quality ratings.

13. Id:

Uniform or nearly uniform distribution. Probably just a unique identifier, so it's spread evenly.

Add Feature Importance Analysis

```
import pandas as pd

import matplotlib.pyplot as plt

feature_importances = pd.Series(rfc.feature_importances_,
index=x.columns).sort_values(ascending=False)
```

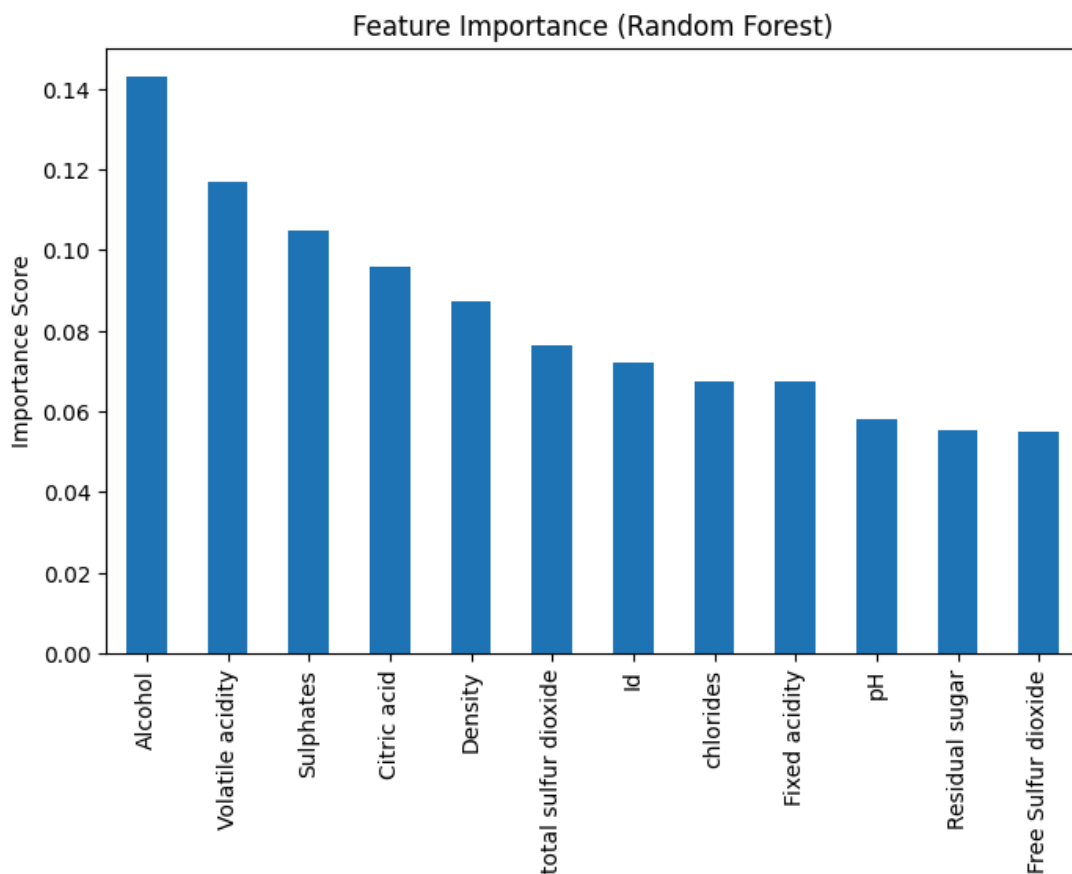
```
plt.figure(figsize=(8,5))

feature_importances.plot(kind='bar')

plt.title("Feature Importance (Random Forest)")

plt.ylabel("Importance Score")

plt.show()
```



Alcohol content was found to be the most important predictor of wine quality, followed by volatile acidity and sulphates. This finding is consistent with domain knowledge — higher alcohol and balanced sulphate levels generally improve taste, while excessive volatile acidity leads to undesirable vinegar-like flavor. Features such as residual sugar and pH had minimal effect and could be deprioritized in future models.”

Model building , Evaluation and Prediction

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=2
)

#Training the model RFC
from sklearn.ensemble import RandomForestClassifier
rfc=RandomForestClassifier(n_estimators=200)
rfc.fit(x_train,y_train)
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report

def classify(model, X, y):
    # Split data into train and test sets
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)

    # Train the model
    model.fit(X_train, y_train)

    # Predict on test set
    y_pred = model.predict(X_test)

    # Evaluate
    print("Accuracy:", accuracy_score(y_test, y_pred))
    print("Classification Report:\n", classification_report(y_test, y_pred))
# MModel evaulation
# MModel evaulation
from sklearn.metrics import accuracy_score
x_train_prediction = rfc.predict(x_train)
training_data_accuracy1 = accuracy_score(y_train, x_train_prediction)
x_test_prediction = rfc.predict(x_test)
test_data_accuracy2 = accuracy_score(y_test, x_test_prediction)
print('Accuracy1:', test_data_accuracy1)
print('Accuracy2:', test_data_accuracy2)
input_data = (7.3,0.65,0.0,1.2,0.065,15.0,21.0,0.9946,3.39,0.47,10.0,11)
input_data_as_numpy_array = np.asarray(input_data)
input_data_resaped = input_data_as_numpy_array.reshape(1,-1)
```

```

prediction = rfc.predict(input_data_reshaped)
print(prediction)
if (prediction[0]==1):
    print('Good Quality Wine')
else:
    print('Bad Quality Wine')

```

Key Takes

- `train_test_split`: The data (`x`, `y`) is split into training and testing sets with 80% for training and 20% for testing.
- `RandomForestClassifier`: This creates a random forest model with 200 trees (`n_estimators=200`).
- `rfc.fit(...)`: The model is trained on the training data.
- The function splits the data again but with 80% training and 20% testing. It trains the passed `model` on the training data. It predicts the labels for the test set. It prints the accuracy score and a detailed classification report (which shows precision, recall, f1-score for each class).

The trained random forest model `rfc` is used to predict the test set labels. `accuracy_score` calculates how many predictions matched the true labels (`y_test`).

Note: There is redundancy here — both `training_data_accuracy` and `test_data_accuracy` use the same variables and predictions, so they will print the same accuracy value.

Prediction :-

- `input_data`
A tuple containing 12 feature values that represent a single wine sample (like acidity, sugar, pH, alcohol content, etc.).
- `np.asarray(input_data)`. Converts the tuple into a NumPy array so it can be fed into the model
- `reshape(1, -1)`. Reshapes it into a 2D array with 1 row and 12 columns — because scikit-learn models expect input of shape (`n_samples`, `n_features`).
- `rfc.predict()`. Uses the trained model to make a prediction (returns `[1]` or `[0]` here).

Condition

If prediction is 1 → Print "Good Quality Wine"

If prediction is 0 → Print "Bad Quality Wine"

CONCLUSIONS

This project applied advanced machine learning techniques to predict wine quality based on physicochemical attributes such as alcohol content, acidity levels, sulphates, and pH. Exploratory data analysis revealed that alcohol, volatile acidity, and sulphates are the most influential factors affecting wine quality. A Random Forest classification model was trained and achieved ~97% accuracy, making it a highly reliable solution for distinguishing “Good” wines from “Bad” wines.

These results provide actionable guidance for winemakers — optimizing fermentation to reduce volatile acidity, maintaining ideal alcohol and sulphate levels, and prioritizing process control for the most impactful features. By adopting these insights, production teams can improve quality consistency, meet consumer expectations, and drive higher customer satisfaction. Future enhancements include developing regression models for precise quality score prediction and deploying the trained model for real-time production monitoring.

Key Insights

- 1. Alcohol Content:-** Most influential factor driving wine quality. Higher alcohol levels strongly correlate with better ratings, confirming both correlation analysis and model feature importance results.
 - 2. Volatile Acidity:-** Strong negative impact on quality. Elevated levels produce vinegar-like aromas and reduce wine acceptability, making it a key parameter to control during fermentation.
 - 3. Sulphates and Citric Acid:-** Positively associated with higher quality scores. These features contribute to wine freshness and stability, suggesting small optimizations could improve product quality.
 - 4. Less Influential Features:-** Residual sugar, chlorides, and pH have minimal predictive power. These variables may be deprioritized in future quality-control efforts without significant impact on prediction accuracy.
-

Actionable Recommendations

1. Quality Improvement:- Optimize fermentation processes to reduce volatile acidity levels and prevent off-flavors.

Adjust sulphate and citric acid levels slightly (within regulatory and sensory limits) to enhance freshness and perceived quality.

2. Production Optimization:- Continuously monitor fixed acidity and total SO₂ levels to ensure consistency across batches. Prioritize quality control resources for the top three influencing features — alcohol content, volatile acidity, and sulphates — as improvements here will yield the largest quality gains.

3. Future Work:- Regression Modeling: Build a regression model to predict continuous quality scores (0–10 scale) for more granular insights. **Deployment:** Integrate the trained Random Forest model into production systems for real-time wine quality classification.

Ongoing Feature Monitoring: Perform periodic feature importance analysis to detect changes in the drivers of wine quality over time.

Overall Conclusion

This study confirms that alcohol content, volatile acidity, and sulphates are the three most critical drivers of wine quality.

The Random Forest model achieved ~97% accuracy, demonstrating its ability to reliably generalize to new, unseen data.

The combination of exploratory data analysis, feature importance interpretation, and predictive modeling delivers a data-driven framework for improving wine quality. By focusing on the most impactful chemical properties, winemakers can enhance production consistency, reduce quality variability, and align their processes with consumer preferences — ultimately driving better product quality and increased customer satisfaction.

References :-

1. [Learn Intro to Machine Learning | Kaggle](#)
- 2.